

Can we lower false positives by moving the threshold?

One saved model · six cutoff rules chosen from mixed validation labels or fixed in advance · no retraining

RULE	AI SCORE CUTOFF	MIXED AI RECALL	MIXED HUMAN FPR	PAPER AI RECALL	PAPER HUMAN FPR	ACL ABSTRACTS HUMAN FPR	PMC BODY HUMAN FPR
Original · 2% validation FPR	3.7%	100.0% 631/631	3.01% 19/631	100.0% 221/221	4.52% 10/221	2.18% 383/17560	5.75% 15/261
At most 1% validation FPR	5.0%	100.0% 631/631	2.22% 14/631	100.0% 221/221	3.62% 8/221	1.96% 344/17560	3.83% 10/261
Zero validation false positives	23.8%	100.0% 631/631	1.27% 8/631	100.0% 221/221	2.26% 5/221	1.13% 199/17560	2.68% 7/261
Standard AI score ≥ 50%	50.0%	100.0% 631/631	0.63% 4/631	100.0% 221/221	0.90% 2/221	0.84% 147/17560	2.68% 7/261
Validation class-gap midpoint	86.4%	100.0% 631/631	0.63% 4/631	100.0% 221/221	0.90% 2/221	0.55% 97/17560	1.53% 4/261
At least 99% validation AI recall	99.992%	98.7% 623/631	0.16% 1/631	98.6% 218/221	0.45% 1/221	0.12% 21/17560	0.00% 0/261

Paper = 221 human + 221 AI abstracts within the mixed test. ACL and PMC body are human-only audits (17,560 and 261 texts).

Cutoffs use the 622-human/622-AI mixed validation set; the 50% rule is fixed. Holdouts were used only to measure outcomes.

Rates are observed on these sets, not a guarantee. Green FPR cells are below 2%. The reported AI score is an uncalibrated softmax value.